

Digital Diabetes Remission Trial (DiGEST)

Statistical Analysis Plan

Version 0.2 from 2023-11-17

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Responsibilities

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Abbreviations

α	Significance level
AE	Adverse events
BMI	Body Mass Index
EDC	Electronic data capture
FAS	Full-analysis-set
FWER	Family-wise error rate
GEE	Generalised estimating equations
HbA1c	Glycated haemoglobin
ITT	Intention-to-treat
MI	Multiple imputation
MICE	Multiple imputation by chained equations
mmHg	Millimetre of mercury
MMRM	Mixed-effects model for repeated measurements
PEP	Primary endpoint
PP	Per-protocol
SAE	Severe adverse events
SEP	Secondary endpoint
SP	Safety population
T2D	Type 2 diabetes mellitus
TDR	Total diet replacement

1 Background

Diabetes remission programmes have recently shown consistently reliable results in patients with type 2 diabetes mellitus (T2D), improving many areas of diabetes control, glycaemia, blood pressure, lipids and microvascular damage. However, such programmes may require intensive face to face visits with Health Care Professionals, resulting in increased costs and low uptake and compliance by patients.

1.1 Trial objective

The aim of this trial is to assess whether diet replacement and a digital therapeutic (Habitual Remission Programme) delivered remotely, leads to remission and weight loss in T2D patients, providing a more cost and resource effective Total Diet Replacement (TDR) remission programme.

1.2 Study design

This study is a prospective, open-label, randomized controlled clinical trial with an active control group, followed by an observational study of the intervention group.

Participants will be assigned to the intervention or control group at a ratio of 2:1, stratified by gender. The intervention group receives the Habitual Remission Programme, which includes 3 months of total diet replacement (TDR) providing 800 kcal/day followed by 6 weeks of diet reintroduction and another 6 weeks of further nutritional education and management of caloric intake, alongside the Habitual digital therapeutic delivered via mobile application. The control group receives no specified treatment and is free to use services in standard usual care. The observational period for the clinical trial is 6 months.

Participants in the intervention group are additionally followed up to 12 months after inclusion in an observational setting.

2 Statistical methods and analytical strategy

The statistical analysis is based on the primary data collected in this study. When evaluating the primary and secondary endpoints, a significance level (α) of 5% is used. For inferential tests, an appropriate effect size measure is calculated in addition to test statistic and p -value.

For inference tests, continuous dependent variables are examined for deviations from the normal distribution using Q-Q plots. If no clear deviations are apparent, parametric procedures are used to test for group differences (t -tests, linear mixed effect models). If significant deviations from the normal distribution are apparent, the appropriate non-parametric procedures are used (Wilcoxon signed-rank tests when comparing paired samples, generalised estimating equations).

2.1 Study populations

Four study populations are constructed for statistical analyses:

- 1) the *Intention-to-treat* population (ITT),

2) the *Full-analysis-set* population (FAS),

3) the *Per-protocol* population (PP),

The ITT population includes all patients who were enrolled into the study and randomised.

2.1.1 FAS

The FAS population includes all patients of the ITT population for whom data on weight, glycated haemoglobin (HbA1c), waist circumference and blood pressure are available at baseline (t_0) and three (t_1) and six months (t_2).

2.1.2 PP

The PP population includes patients in the FAS population who have adhered to the treatment protocol at three months (t_1).

2.1.3 SP

The SP includes patients in the ITT population who have received at least one TDR “meal”.

2.2 Variables

The following variables are collected during the study. Table 1 displays the measurement timepoints of each variable for each outcome.

Table 1: Variables and measurement timepoints

		Measurement time point			
		Baseline	3 months	6 months	12 months (intervention group only)
Outcome/variable	Outcome measure / operationalisation	t_0	t_1	t_2	t_3
Weight	Weight in kg	X	X	X	X
HbA1c	HbA1c in relative (%) and absolute units (mmol/mol)	X		X	X
Waist circumference	Waist circumference in cm	X	X	X	X
Blood pressure	Systolic blood pressure in mmHg	X	X	X	X
	Diastolic blood pressure in mmHg	X	X	X	X
Medication use	Use of glucose-lowering medication	X	X	X	X
Safety	Number of AE	Continuous assessment			
	Number of SAE	Continuous assessment			

	Number of (S)AEs that constitute Major Adverse Cardiovascular Events and Major Adverse Diabetes Events	Continuous assessment			
Adherence	Number of participants completing the intervention		X	X	

2.2.1 Primary endpoint

The primary endpoint is reaching a weight loss of at least 15kg at 6 months compared to baseline ($\text{weight}_{t_2} - \text{weight}_{t_0}$), or reaching an HbA1c-value of less than 6.5% (<48 mmol/mol) at 6 months (t_2), after at least 2 months off all glucose-lowering medication.

2.2.2 Secondary endpoints

The study assesses 11 secondary endpoints, which are summarised below:

1. Reduction of **HbA1c between t_0 , t_1 , and t_2** ,
2. Change in **weight between t_0 , t_1 , and t_2** ,
3. Change in **waist circumference between t_0 , t_1 , and t_2** ,
4. Change in **systolic blood pressure between t_0 , t_1 , and t_2** ,
5. Change in diastolic **blood pressure between t_0 , t_1 , and t_2** ,
6. Change in self-reported **number of glucose-lowering substances used between t_0 , t_1 , and t_2** ,
7. **Weight loss of at least 5% (10%, 15%)** at 6 months compared to baseline ($(\text{weight}_{t_2} - \text{weight}_{t_0})/\text{weight}_{t_0}$)
8. **Achieving an HbA1c-value of less than 6.5% (<48 mmol/mol)** at 6 months (t_2) after at least 2 months off all glucose-lowering medication
9. Investigating the **adherence to the intervention**, assessed by the number of participants completing the intervention at 3 (t_1) and 6 months (t_2)
10. Evaluating the **safety of the intervention**, assessed by reported AEs, SAEs, and the number of (S)AEs that constitute Major Adverse Cardiovascular Events and Major Adverse Diabetes Events between baseline (t_0) and 6 months (t_2)
11. **Weight loss of at least 5% (10%, 15%)** at 12 months compared to baseline ($(\text{weight}_{t_3} - \text{weight}_{t_0})/\text{weight}_{t_0}$)
12. **Achieving an HbA1c-value of less than 6.5% (<48 mmol/mol)** at 12 months (t_3) after at least 2 months off all glucose-lowering medication

2.2.3 Patient demographics and baseline characteristics

The following information on demographic and other descriptive characteristics of the patient population will be collected:

- Age, sex, weight, height Body Mass Index (BMI)

2.3 Missing values

Missing values in outcome variables, covariates, or instrumental variables are imputed using multiple imputation (MI) procedures specifically, the *Multiple Imputation by Chained Equations* (MICE) technique is used (Buuren and Groothuis-Oudshoorn 2011). A number of 5 imputed datasets will be generated and used for the analysis. Results will be pooled according to Rubin's rule.

The number of missing values in the outcome and descriptive variables is reported at each follow-up point.

2.4 Statistical Methods

The following paragraph describes the statistical methods used to summarise and describe the data, as well as the statistical tests to assess the primary and secondary endpoints. For statistical hypothesis testing, effect sizes will be reported alongside p -values, and p -values < 0.05 will be considered statistically significant.

2.4.1 Descriptive Statistics

Descriptive statistical measures are used to summarize the endpoints, covariates, and instrumental variables at the follow-up points. The frequency of missing values per variable and study time point is reported, but missing values are not replaced.

For categorical variables, absolute and relative frequencies are reported. For ordinal variables, median, minimum and maximum values and quartiles are used, for metric variables, arithmetic means and standard deviations are reported additionally. The distribution of the outcome variables is also displayed via boxplots or equivalent graphical procedures for each group and follow-up point. Measures of central tendency are displayed with variability parameters (e.g., arithmetic means with standard deviation/standard error/confidence interval).

2.4.2 Primary endpoint

The primary endpoint is reaching a weight loss of at least 15kg at 6 months compared to baseline ($\text{weight}_{t_2} - \text{weight}_{t_0}$), or reaching an HbA1c value of less than 6.5% (<48 mmol/mol) at 6 months (t_2), after at least 2 months off all glucose-lowering medication.

To control the family-wise error rate (FWER), p -values are adjusted for multiple comparisons using the Bonferroni-Holm method (Holm 1979). First, the p -values are ordered from smallest to largest. The smallest p -value is then compared against the threshold of $\alpha = 5\%/m$, where m is the number of comparisons. The next smallest p -value is then compared against a threshold of $\alpha = 5\%/(m-1)$, and

so on, until the final p-value will be compared against the threshold $\alpha = 5\%$. This approach ensures a total FWER of 5% across all comparisons.

The likelihood of reaching a weight loss of at least 15kg or reaching an HbA1c value of less than 6.5% will be compared between groups using two logistic regression models, one for each outcome, with the following variables:

Dependent variable:

- Weight loss of at least 15kg [HbA1c value less than 6.5%] (yes/no)

Independent variables/covariates:

- Weight [HbA1c value] at t_0
- Group membership
- Sex

All analyses for the primary endpoints are carried out using the ITT population. If more than 10% of the data required to calculate the primary endpoint are missing, a sensitivity analysis will be conducted using imputed values.

2.4.3 Secondary endpoints

The secondary objectives will be analysed using the following strategy, described for each endpoint below. The ITT population will be used for analysis. If more than 10% of the data required to calculate the respective endpoint are missing, a sensitivity analysis will be conducted using imputed values.

Change in HbA1c, weight, waist circumference, systolic blood pressure, diastolic blood pressure between t_0 , t_1 , and t_2

The outcome at each time point will be evaluated for its distribution using Q-Q plots. If the data appear to follow a normal distribution, a mixed effects model for repeated measurements (MMRM) is used to analyse the data. If large deviations from the normal distribution are apparent, generalised estimating equations (GEE) are used. The following variables enter the model:

Dependent Variable:

- Outcome

Independent variables/covariates:

- Group membership
- Measurement point in time
- Outcome score at t_0
- Sex

Change in self-reported number of glucose-lowering substances used between t_0 , t_1 , and t_2

At each follow up point in time, self-reported number of glucose-lowering substances used is recorded. A Poisson MMRM is used to analyse the data. The following variables enter the model:

Dependent Variable:

- Number of glucose-lowering substances used

Independent variables/covariates:

- Group membership
- Measurement point in time
- Sex

Weight loss of at least 5% (10%, 15%) at 6 months compared to baseline ((weight_{t2} - weight_{t0})/weight_{t0})

This endpoint is analysed using a logistic regression model with the following variables:

Dependent Variable:

- Weight loss of at least 5% (10%, 15%) compared to baseline (yes/no)

Independent variables/covariates:

- Group membership
- Weight at t_0
- Sex

Achieving an HbA1c-value of less than 6.5% at 6 months after at least 2 months off all glucose lowering medication

This endpoint is analysed using a logistic regression model with the following variables:

Dependent Variable:

- HbA1c-value of less than 6.5% (yes/no)

Independent variables/covariates:

- Group membership
- HbA1c-value at t_0
- Sex

Investigating the adherence to the intervention, assessed by the number of participants completing the intervention at 3 (t_1) and 6 months (t_2)

The absolute and relative number of participants completing the intervention at 3 (t_1) and 6 months (t_2) is analysed descriptively for the intervention group.

Evaluating the safety of the intervention, assessed by reported AEs, SAEs, and the number of (S)AEs that constitute Major Adverse Cardiovascular Events and Major Adverse Diabetes Events between baseline (t_0) and 6 months (t_2)

The absolute and relative frequency of AEs and SAEs and the number of (S)AEs that constitute Major Adverse Cardiovascular Events and Major Adverse Diabetes Events between baseline (t_0) and 6 months (t_2) and 12 months (t_3 , intervention group only) is reported for each group using the ITT population.

Weight loss of at least 5% (10%, 15%) at 12 months compared to baseline (($weight_{t_3} - weight_{t_0}$)/ $weight_{t_0}$)

This endpoint is analysed using a logistic regression model with the following variables:

Dependent Variable:

- Weight loss of at least 5% (10%, 15%) compared to baseline (yes/no)

Independent variables/covariates:

- Group membership
- Weight at t_0
- Sex

Achieving an HbA1c-value of less than 6.5% at 12 months after at least 2 months off all glucose lowering medication

This endpoint is analysed using a logistic regression model with the following variables:

Dependent Variable:

- HbA1c-value of less than 6.5% (yes/no)

Independent variables/covariates:

- Group membership
- HbA1c-value at t_0
- Sex

2.4.4 Sensitivity analyses

Sensitivity analyses are performed to determine whether assumptions or specifications made in the preparation and analysis of the data had an influence on the effects of the primary and secondary

endpoints. In this study, this concerns the choice of the analysis population and handling missing values. The analyses for the primary and endpoints will be repeated with different study populations as appropriate. The planned sensitivity analyses are summarized in Table 2. Additionally, a sensitivity analysis will re-assess the primary endpoint using imputed values if more than 10% of the data required to calculate the primary endpoint are missing.

Table 2: Summary of planned sensitivity analyses

No.	Endpoint	Variable	Population	Covariates
1	PEP	Weight reduction of at least 15kg from t_0 to t_2 or HbA1c of less than 6.5% at t_2	FAS	See main analysis
2	PEP	Weight reduction of at least 15kg from t_0 to t_2 or HbA1c of less than 6.5% at t_2	PP	See main analysis
3	SEP1	HbA1c reduction between t_0 and t_2	PP	See main analysis
4	SEP2	Weight change between t_0 and t_2	PP	See main analysis
5	SEP3	Waist circumference change between t_0 and t_2	PP	See main analysis
6	SEP4	Systolic blood pressure change between t_0 and t_2	PP	See main analysis
7	SEP5	Diastolic blood pressure change between t_0 and t_2	PP	See main analysis
8	SEP6	Change in use of glucose-lowering medication between t_0 and t_2	PP	See main analysis
9	SEP7	Weight reduction of at least 5% (10%, 15%) from t_0 to t_2	PP	See main analysis
10	SEP8	HbA1c of less than 6.5% at t_2	PP	See main analysis

2.4.5 Subgroup analyses

Subgroup analyses will examine whether effects differ descriptively between predefined groups. In this study, analyses of the primary and secondary endpoints are planned separately according to the participant's sex.

2.5 Sample size determination

Sample size calculations are based on the remission rates of DiRECT at 12 months (46% in the intervention group and 4% in the control). Conservative calculations have assumed diabetes remission in 32% of intervention participants at 6 months compared with 4% in the control group. Allowing for an estimated 25% individual participant drop-out within 6 months, we will recruit a total of 100 participants to provide at least 90% power to detect significant differences at an α of 5%. Participants will be recruited in cohorts using a 2:1 ratio (intervention:control).

3 Protocol deviations

Missing more than 14 consecutive days of meals during the TDR phase will be considered a violation of treatment protocol. Adherence to the treatment protocol will be assessed through telephone surveys at the end of the TDR phase. Participants who reported a treatment protocol violation will be excluded from the PP population.

4 Data management

Data will be entered into a validated electronic data capture (EDC) platform. Direct access to the EDC platform will be granted to authorised representatives from the Sponsor, Lindus Health and the regulatory authorities to allow trial-related monitoring, audits and inspections. The data collected as part of the clinical study are pseudonymised and stored in an encrypted database that is technically secured against unauthorized access. A list containing pseudonyms and identifiable information is kept separately from the study data.

5 Software

Data analysis is performed using the software *R* in Version 4.2.2 or later.

6 References

- Buuren, Stef van, and Karin Groothuis-Oudshoorn. 2011. 'Mice: Multivariate Imputation by Chained Equations in R'. *Journal of Statistical Software* 45 (December): 1–67. <https://doi.org/10.18637/jss.v045.i03>.
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